

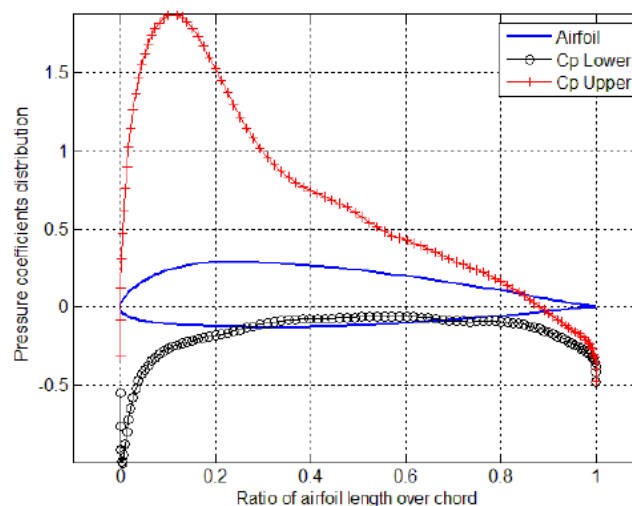
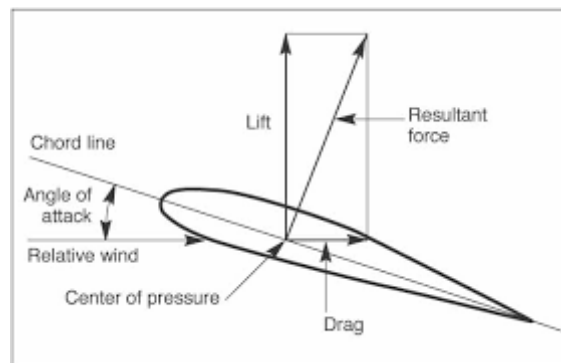
Drag Force

Drag: Resistance force overcome by the thrust force dictates the efficiency of the aircraft.

- Pressure and viscous force lead to drag on the airfoil.
- Drag → Pressure Imbalance + Viscous Effect
- Flow Separation → Root cause for the pressure drag.

All the elementary flows are inviscid and incompressible and can be used to estimate the pressure coefficients.

Pressure coefficients → Lift and drag (only based on pressure coefficient, viscous effect (shear stress) ignored)



We know that,

$$L = N \cos \alpha - A \sin \alpha$$

$$D = N \sin \alpha + A \cos \alpha$$

Where N is normal force and A is axial force.

Panel methods → Accurately predict the Lift.

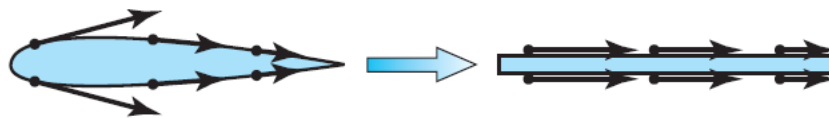
Assuming **Invalid** → Not allowing for flow separation (ignored all drag-causing mechanisms).

Viscous drag:

- Drag due to **viscous force**.
- **Viscous force**: comes from the bouncing of fluid molecules from one streamline to the other causing change in momentum (**Rate of change of momentum = Force, $F = ma$**).
- Velocity gradient inside the boundary layer → Shear stress on the surface.

Approximation:

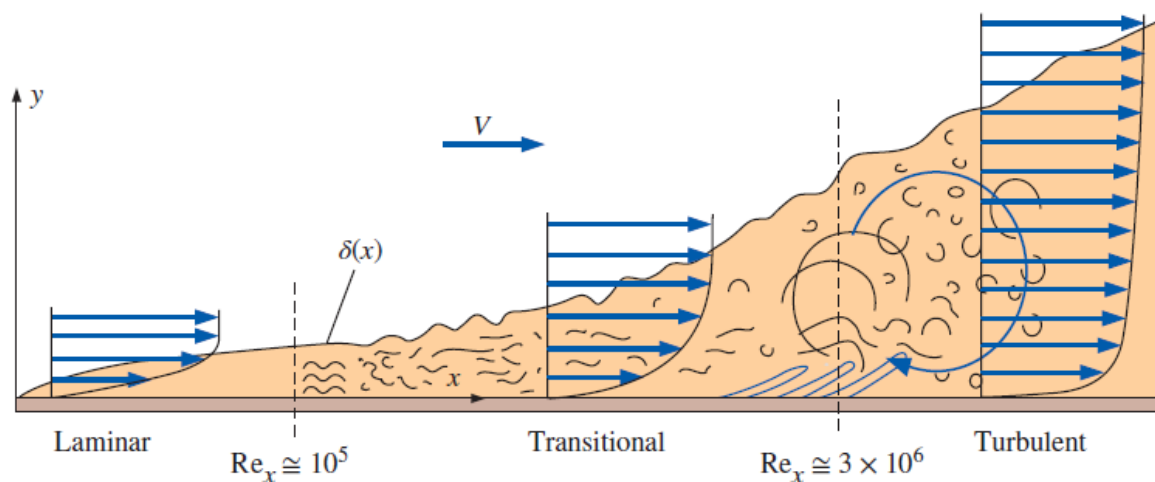
If the airfoil is thin and at a small angle of attack, it can be approximated as a flat plate.



We know a lot about the development of boundary layers over a flat plate.

Curved surface – There are pressure gradients on the curved surface that makes harder to derive the analytical expressions for boundary layer similar to Flat place.

Flat Plate Boundary Layer:



- Flow passes over a flat plate the boundary layer grows once it comes into contact with the plate.
- Laminar BL → Grows parabolically.
- At some distance down the plate which is at a critical x location, the BL rapidly grows and the flow transitions to turbulence.

- Turbulent flow → Unsteady, chaotic, and high energy.
- Both the laminar and turbulent BL have velocity gradient at the wall and produce shear stress distribution that vary with x.
- This shear stress distribution integrates to a total drag on the plate.

Summary of expressions for laminar and turbulent boundary layers on a smooth flat plate aligned parallel to a uniform stream*

Property	Laminar	(a)	(b)
		Turbulent ^(†)	Turbulent ^(‡)
Boundary layer thickness	$\frac{\delta}{x} = \frac{4.91}{\sqrt{Re_x}}$	$\frac{\delta}{x} \cong \frac{0.16}{(Re_x)^{1/7}}$	$\frac{\delta}{x} \cong \frac{0.38}{(Re_x)^{1/5}}$
Displacement thickness	$\frac{\delta^*}{x} = \frac{1.72}{\sqrt{Re_x}}$	$\frac{\delta^*}{x} \cong \frac{0.020}{(Re_x)^{1/7}}$	$\frac{\delta^*}{x} \cong \frac{0.048}{(Re_x)^{1/5}}$
Momentum thickness	$\frac{\theta}{x} = \frac{0.664}{\sqrt{Re_x}}$	$\frac{\theta}{x} \cong \frac{0.016}{(Re_x)^{1/7}}$	$\frac{\theta}{x} \cong \frac{0.037}{(Re_x)^{1/5}}$
Local skin friction coefficient	$C_{f,x} = \frac{0.664}{\sqrt{Re_x}}$	$C_{f,x} \cong \frac{0.027}{(Re_x)^{1/7}}$	$C_{f,x} \cong \frac{0.059}{(Re_x)^{1/5}}$

How good is the approximation (Airfoil → Flat plate)?

NACA 2412

At Re = 3.1 million for the chord length of 1.5 m

Drag coefficient = 0.0068

Fully Laminar BL over the plate:

Drag coefficient = 0.0015 (**Under-estimated**)

Fully Turbulent BL over the plate:

Drag coefficient = 0.00794 (**Over-estimated**)

In reality, the flow naturally transitions from Laminar to Turbulent.

The transition happens at a significant x value which we need in order to accurately get the drag.

Key parameter: Critical Reynolds number, $Re_{cr} = \rho U_{\infty} x_{cr} / \mu$

Estimating critical location → Theoretically impossible (can be estimated from experience, measurements, simulations)

Laminar + Turbulent BL over the plate:

For a flat plate, $Re_{cr} = 5 \times 10^5$ (approximately)

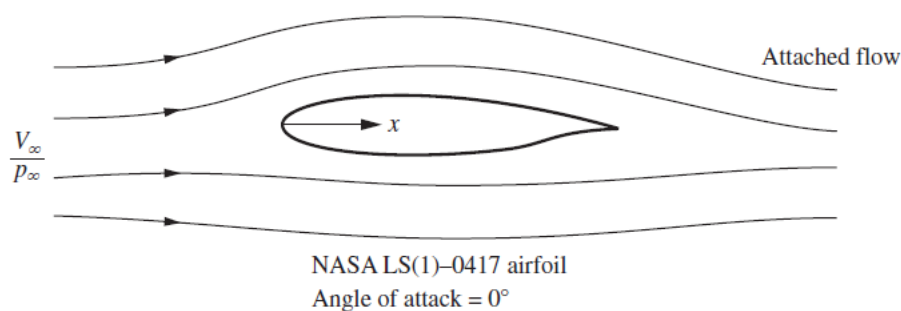
Drag coefficient = 0.0062 (close to the real value but not accurate because pressure drag is ignored)

NACA 2412: Drag coefficient = 0.0068 (Viscous drag coefficient + Pressure drag coefficient)

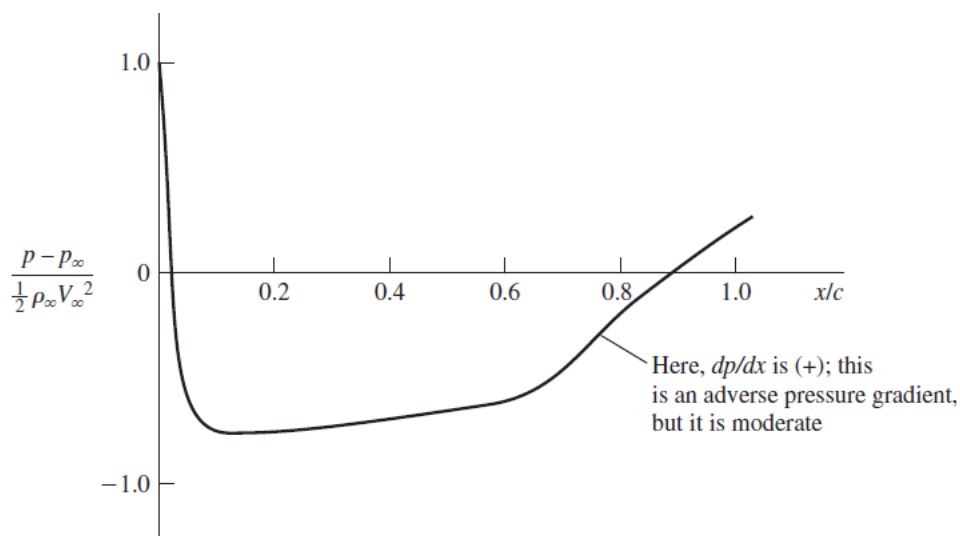
Pressure drag:

- **Imbalance of pressure due to separation.**

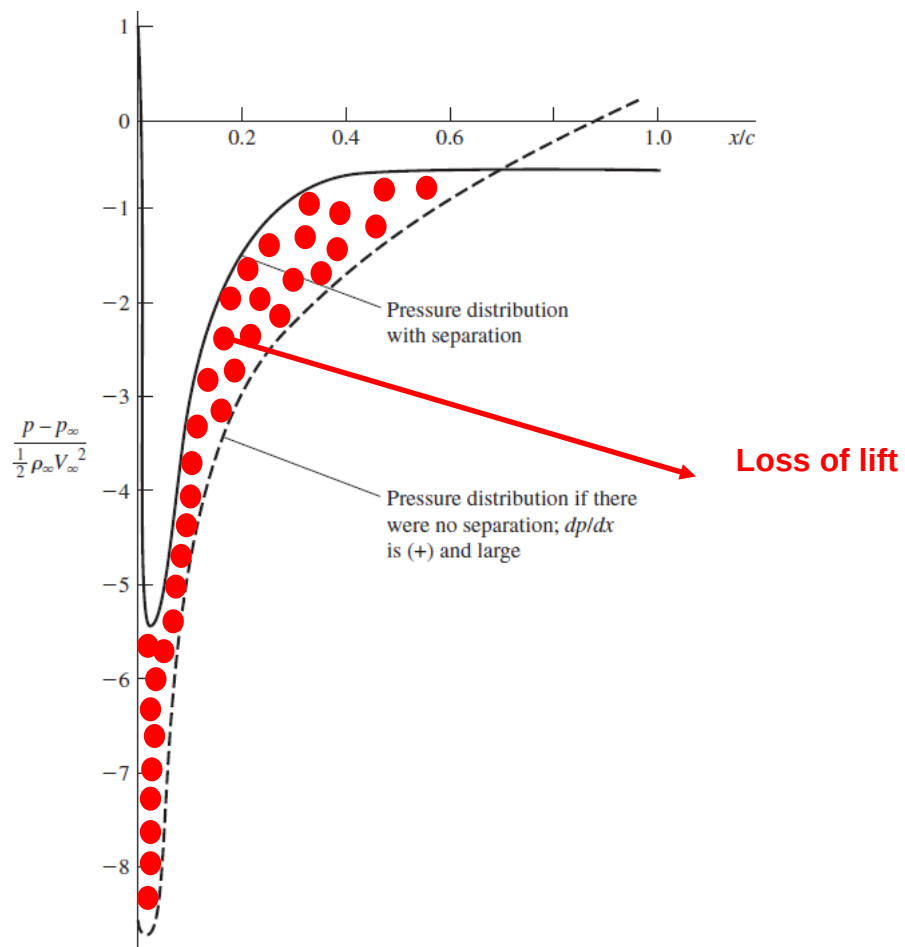
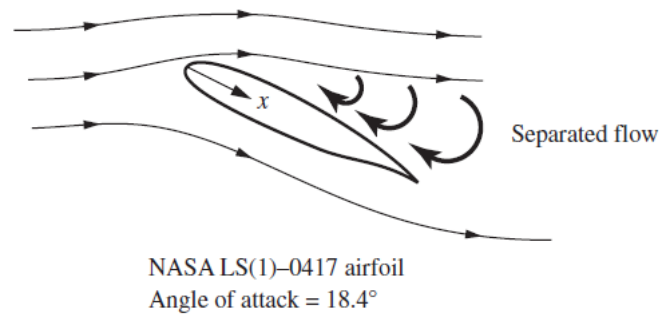
Consider flow over a NASA low-speed airfoil,



As flow passes over the nose and up the foil, it accelerates and experience a **favourable** pressure gradient (**dp/dx – Negative**).



At high angle of attack,



Therefore, two major consequences of the flow separating over an airfoil are:

1. A drastic loss of lift (stalling).
 2. A major increase in drag, caused by pressure drag due to flow separation.
- Flow separation causes higher pressure on the upper surface (compared to the attached flow) as shown in figure (below).
 - Pressure at the trailing edge due to separation is lower than the attached flow (**causes pressure imbalance leads to net force, Pressure force**)

